

# Florence-2: Advancing a Unified Representation for a Variety of Vision Tasks

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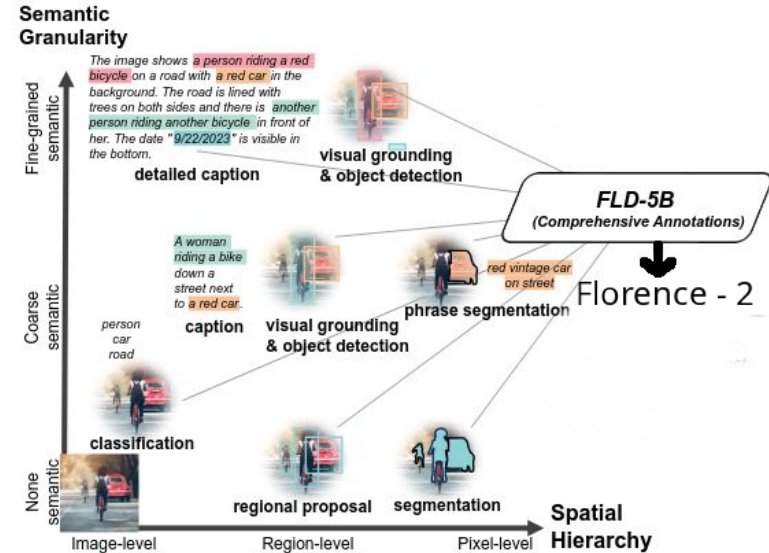
# Introduction

## What is Florence-2?

- A vision foundation model developed by Microsoft.
- Implements a unified, prompt-based representation for computer vision and vision-language tasks.
- Handles multiple vision tasks including captioning, object detection, segmentation, and grounding.
- Uses text prompts to execute tasks rather than separate architectures for different tasks.

## Why is Florence-2 important?

- Bridges the gap between task-specific models and generalized vision models.
- Overcomes data limitations and lack of unified architectures in vision models.
- Achieves state-of-the-art zero-shot and fine-tuned performance.



# Motivation

## Why Do We Need Florence-2?:

- Existing models struggle with diverse vision tasks when using simple instructions.
- No unified framework to handle spatial hierarchy and semantic granularity simultaneously.
- Florence-2 provides a generalized vision model capable of adapting across multiple vision tasks.

## Key Contributions:

- Uses multitask learning with large-scale annotations.
- Provides a task-agnostic approach to vision modeling.
- Bridges the gap between vision and language understanding.

# Problem Statement & Challenges

**Problem Statement:** Existing vision models struggle with handling diverse tasks using a unified framework due to limitations in scalability, adaptability, and comprehensive annotations, requiring a new approach that integrates spatial hierarchy and semantic granularity into a single architecture.

## Challenges :

1. Limited scalability & efficiency in handling multiple vision tasks.
2. Task-specific models require independent fine-tuning, leading to increased computational cost.
3. Data limitations: Existing datasets lack comprehensive labeled annotations.
4. No single model integrates spatial hierarchy and semantic granularity effectively.

## Florence-2's Solution:

- Provides a unified model for multiple vision tasks.
- Leverages large-scale annotated datasets to improve generalization.
- Uses a sequence-to-sequence learning framework to maintain flexibility across tasks.

# Existing Methods and Limitation

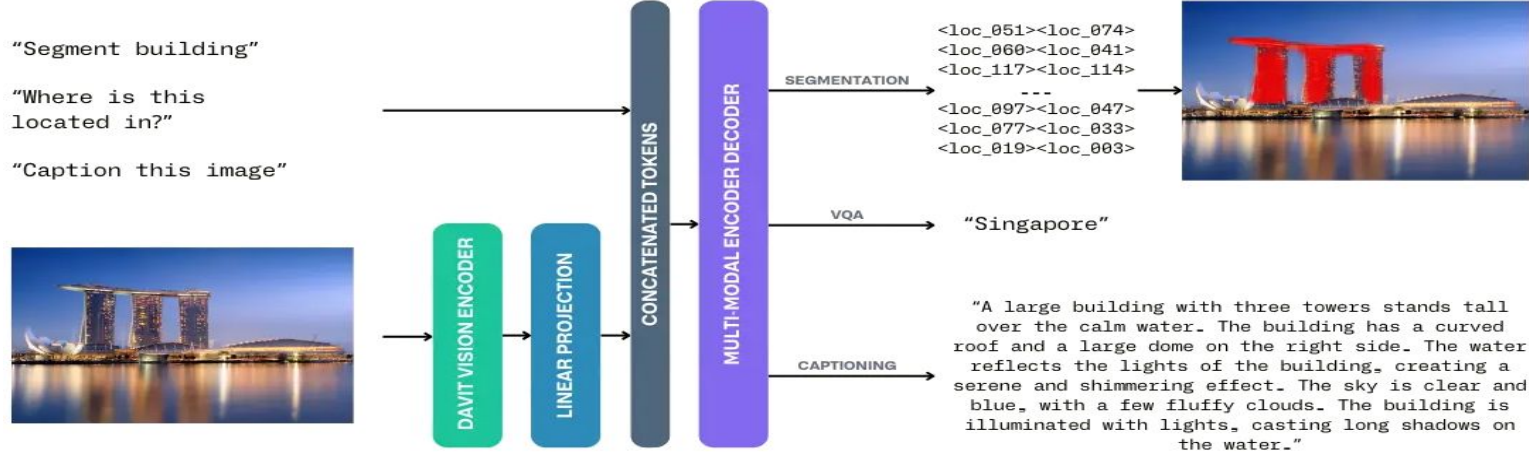
## Traditional Approaches and Their Drawbacks:

1. **Supervised Learning** (ImageNet, COCO) – Strong at object recognition, lacks adaptability.
2. **Self-Supervised Learning** (SimCLR, MAE) – Captures deep features but struggles with fine control.
3. **Weakly Supervised Learning** (CLIP, Florence-1, SAM) – Limited in spatial understanding.

## Key Models Compared:

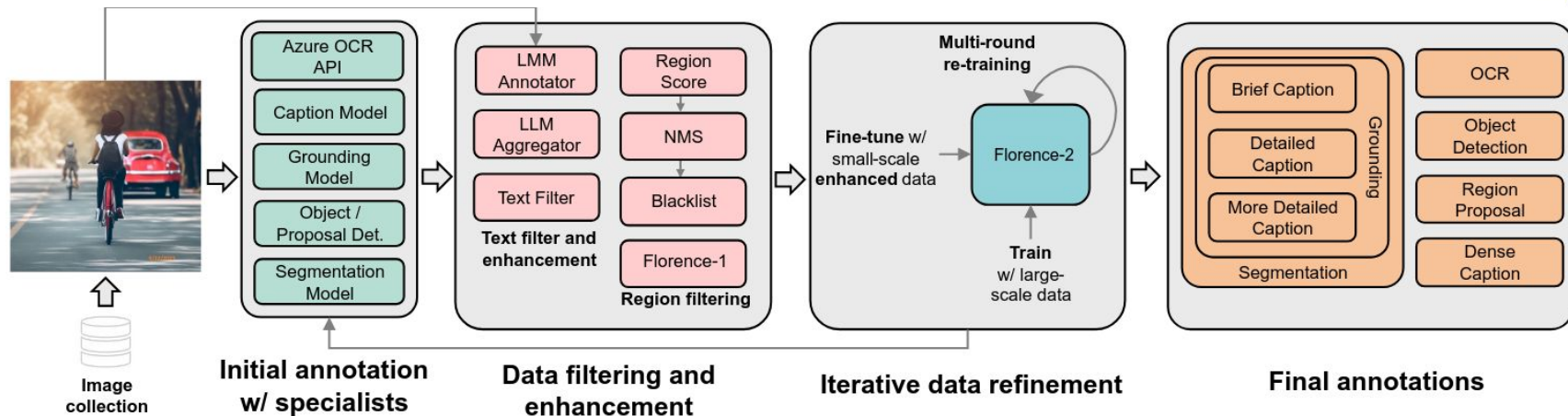
| Model      | Training Type      | Strengths                     | Limitations                  | Key Performance Metrics                                            |
|------------|--------------------|-------------------------------|------------------------------|--------------------------------------------------------------------|
| CLIP       | Weakly Supervised  | Strong image-text alignment   | Limited spatial reasoning    | Zero-shot ImageNet Accuracy: 76.2% with ViT-L/14-336px model.      |
| DINO       | Self-Supervised    | Effective object detection    | Lacks multitask adaptability | COCO Object Detection mAP: 56.9% with DINO ResNet-50 model.        |
| Florence-2 | Multitask Learning | Unified vision language model | Requires large computing     | COCO Captioning CIDEr Score: 135.6 with the large model (0.77B P). |

# Methodology (Florence-2 Approach)



- Unified Vision-Language Model: Handles multiple vision tasks using text prompts instead of separate task-specific architectures.
- Sequence-to-Sequence (seq2seq) Framework:
  1. Uses an image encoder to extract features.
  2. A multi-modality transformer processes both text and image embeddings.
  3. Generates task-specific text-based outputs (e.g., captions, object locations).
- Text-Prompt Adaptation: Enables zero-shot learning across vision tasks without task-specific tuning & Uses extensive pretraining on FLD-5B, making it highly adaptable.

# Methodology (FLD-5B Dataset)



## Largest Dataset Used for Training Florence-2

Size: 126M images, 5.4B annotations – largest dataset for multitask vision pretraining.

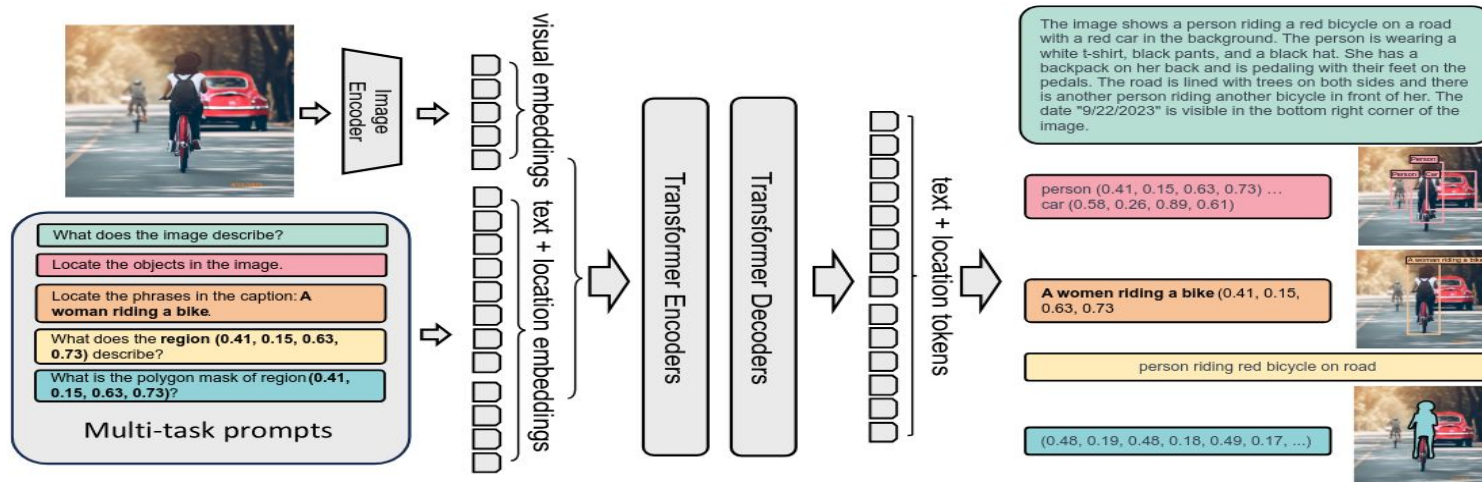
- Annotations cover multiple levels of detail:
  - Image-level (Captions, Classifications)
  - Region-level (Object Detection, Grounding)
  - Pixel-level (Segmentation, Phrase Alignment)

## Why is FLD-5B Important?

- Overcomes data scarcity by generating large-scale, high-quality annotations.
- Fully automated annotation pipeline ensures efficiency & consistency.



# Methodology (Model Architecture)



## 1. Florence-2 Model Components

| Component                                      | Description                                                                                                                                     |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Vision Encoder (DaViT – Dual Attention ViT)    | Extracts visual token embeddings from images, handling spatial hierarchy efficiently. Weights are initialized from UniCL.                       |
| Multi Modality Transformer (Encoder - Decoder) | Processes both text & visual embeddings. Uses cross-attention layers to align vision-language understanding. Weights are initialized from BART. |
| Text-Based Output Generation                   | Generates captions, object locations, segment masks, and other results in a text format. Uses a unified loss function for optimization.         |

## 2. Unified Multi-Task Learning Approach

- FLD-5B dataset annotations are converted into text, making all tasks trainable under a single objective, so No need for separate model heads tasks are controlled via text prompts.
- Optimization: The model uses a cross-entropy loss function:

$$L = - \sum_{i=1}^{|y|} \log P_{\theta}(y_i | y_{<i}, x)$$

- Florence-2's seq2seq framework enables task-agnostic adaptability.
- Unified vision-language processing improves scalability and efficiency.

## 3. Florence-2 Model Variants

| Model Size   | Image Encoder (DaViT) | Encoder-Decoder     | #Params |
|--------------|-----------------------|---------------------|---------|
| Florence-2-B | 90M params            | 6 layers, 6 heads   | 232M    |
| Florence-2-L | 360M params           | 12 layers, 12 heads | 771M    |

# Experimental Setup

## 1. Training Setup & Computational Resources

- Florence-2 was trained on 512 NVIDIA A100 GPUs over 10 days.
- Uses FLD-5B dataset, providing 5.4B high-quality annotations across 126M images.
- Pretraining: Conducted on a large-scale dataset with multiple vision tasks in a single learning framework.

## 2. Multi-Task Learning & Pretraining Approach

- Training follows a unified multi-task learning strategy:
  - a. Uses standardized textual outputs across different tasks.
  - b. Tasks include captioning, object detection, segmentation, grounding, and classification.
- Loss Function: Florence-2 uses a cross-entropy loss function for task optimization.
- Optimization Algorithm: Standard AdamW optimizer with weight decay.

## 3. Evaluation Datasets Used

- Image Captioning & Object Detection: COCO
- Visual Grounding: Flickr30k, RefCOCO, RefCOCO+
- Segmentation: RefCOCOg
- Classification: ImageNet

# Experimental Setup

## 4. Testing Process & Metrics Used

- **Zero-Shot Testing:** Florence-2 was evaluated without additional fine-tuning to test generalization ability.
- **Fine-Tuned Testing:** The model was trained further on public human-annotated datasets for specialized evaluation.
- **Evaluation Metrics Used:**
  - Image Captioning (COCO Captioning) → Model evaluates descriptive accuracy (CIDEr, BLEU).
  - Object Detection (COCO Detection) → Measures object localization accuracy (mAP).
  - Visual Grounding (Flickr30k, RefCOCO) → Tests model's ability to locate objects based on text queries (Recall@1).
  - Segmentation (RefCOCOg) → Evaluates pixel-wise segmentation accuracy (mIOU).

## 5. Key Findings from Testing

- Florence-2 outperformed previous models (CLIP, DINO, UNINEXT, Flamingo) in zero-shot tasks.
- Achieved state-of-the-art performance in object detection, grounding, and segmentation.
- Handles multiple vision tasks efficiently without requiring task-specific models.

# Results

| Method              | #params | COCO Cap.     | NoCaps       | TextCaps     | COCO Det.      | Flickr30k   | Refcoco |        |        | Refcoco+ |        |        | Refcog |      | Refcoco RES |
|---------------------|---------|---------------|--------------|--------------|----------------|-------------|---------|--------|--------|----------|--------|--------|--------|------|-------------|
|                     |         | test<br>CIDEr | val<br>CIDEr | val<br>CIDEr | val2017<br>mAP | test<br>R@1 | val     | test-A | test-B | val      | test-A | test-B | val    | test | val         |
| Flamingo [3]        | 80B     | 84.3          | -            | -            | -              | -           | -       | -      | -      | -        | -      | -      | -      | -    | -           |
| Kosmos-2 [48]       | 1.6B    | -             | -            | -            | -              | 78.7        | 52.3    | 57.4   | 47.3   | 45.5     | 50.7   | 42.2   | 60.6   | 61.7 | -           |
| <i>Florence-2-B</i> | 0.23B   | 133.0         | 118.7        | 70.1         | 34.7           | 83.6        | 53.9    | 58.4   | 49.7   | 51.5     | 56.4   | 47.9   | 66.3   | 65.1 | 34.6        |
| <i>Florence-2-L</i> | 0.77B   | 135.6         | 120.8        | 72.8         | 37.5           | 84.4        | 56.3    | 61.6   | 51.4   | 53.6     | 57.9   | 49.9   | 68.0   | 67.0 | 35.8        |

## Key Analysis for Zero-Shot Performance

- Florence-2 significantly outperforms Flamingo & Kosmos-2 in captioning (COCO, TextCaps).
- Achieves higher object detection accuracy (COCO Detection mAP: 37.5 vs. 34.7 in smaller model).
- Flickr30k retrieval score (R@1: 84.4) is best among tested models.
- Strong segmentation results in RefCOCO RES (mIoU: 35.8).

# Results

| Method            | #params | COCO Det.      | Flickr30k   | RefCOCO |                    |        | RefCOCO+ |                    |        | RefCOCOg |                  | RefCOCO RES |
|-------------------|---------|----------------|-------------|---------|--------------------|--------|----------|--------------------|--------|----------|------------------|-------------|
|                   |         | val2017<br>mAP | test<br>R@1 | val     | test-A<br>Accuracy | test-B | val      | test-A<br>Accuracy | test-B | val      | test<br>Accuracy | val<br>mIoU |
| Specialist Models |         |                |             |         |                    |        |          |                    |        |          |                  |             |
| SeqTR [76]        | -       | -              | -           | 83.7    | 86.5               | 81.2   | 71.5     | 76.3               | 64.9   | 74.9     | 74.2             | -           |
| PolyFormer [42]   | -       | -              | -           | 90.4    | 92.9               | 87.2   | 85.0     | 89.8               | 78.0   | 85.8     | 85.9             | 76.9        |
| UNINEXT [65]      | 0.74B   | 60.6           | -           | 92.6    | 94.3               | 91.5   | 85.2     | 89.6               | 79.8   | 88.7     | 89.4             | -           |
| Ferret [68]       | 13B     | -              | -           | 89.5    | 92.4               | 84.4   | 82.8     | 88.1               | 75.2   | 85.8     | 86.3             | -           |
| Generalist Models |         |                |             |         |                    |        |          |                    |        |          |                  |             |
| UniTAB [66]       | -       | -              | -           | 88.6    | 91.1               | 83.8   | 81.0     | 85.4               | 71.6   | 84.6     | 84.7             | -           |
| Florence-2-B      | 0.23B   | 41.4           | 84.0        | 92.6    | 94.8               | 91.5   | 86.8     | 91.7               | 82.2   | 89.8     | 82.2             | 78.0        |
| Florence-2-L      | 0.77B   | 43.4           | 85.2        | 93.4    | 95.3               | 92.0   | 88.3     | 92.9               | 83.6   | 91.2     | 91.7             | 80.5        |

## Key Analysis for Fine-Tuned Performance

- Specialist Models → Fine-tuned on individual tasks, leading to higher accuracy but task-specific limitations.
- Generalist Models → Performs well across multiple tasks using a single architecture.
- mAP (COCO Detection): 43.4% (vs. UNINEXT 60.6% at 0.74B params).
- RefCOCO+ (test A/B Accuracy): 88.3% / 92.9% (competitive with specialist models).
- Segmentation (mIOU on RefCOCO RES): 80.5% (better than some specialist models).
- Florence-2 achieves near-specialist accuracy despite being a generalist model.

# Limitations

## High Computational Cost

- Training required High GPU Computation (Multiple A100 GPUs).
- Deployment in low-resource environments is challenging.

## Dataset Bias

- FLD-5B dataset is web-crawled and may contain biases in annotations.
- Model performance may vary across underrepresented datasets.

## Zero-Shot Limitations in Some Tasks

- While Florence-2 performs well in zero-shot vision tasks, It still lags behind specialist models in highly domain-specific applications.
- Fine-tuning improves performance, but task adaptation remains a challenge.

## Potential Vulnerabilities to Adversarial Attacks

- Like other large-scale models, Florence-2 may be susceptible to adversarial perturbations.
- Needs robust defense mechanisms for real-world applications.

# Conclusion

- **Unified Vision-Language Model:** Florence-2 integrates multiple vision and vision-language tasks (captioning, object detection, segmentation, and grounding) into a single, prompt-based sequence-to-sequence (seq2seq) framework, eliminating the need for task-specific architectures.
- **Massive-Scale Training & Data Efficiency:** Trained on the FLD-5B dataset with 5.4 billion annotations, it leverages large-scale supervision to achieve state-of-the-art performance in both zero-shot and fine-tuned settings.
- **Model Efficiency & Competitive Performance:** Despite being significantly smaller than Flamingo (80B) and Kosmos-2 (1.6B), Florence-2 delivers comparable or superior results, demonstrating high efficiency in multi-task vision processing.
- **Challenges & Future Considerations:** While highly scalable, challenges such as high computational demands, dataset biases, and susceptibility to adversarial attacks remain critical areas for future improvements and deployment considerations.



# Plan of Novelty

## Why Study Adversarial Attacks?

- Vision models are vulnerable to adversarial perturbations, leading to incorrect predictions.
- Even small modifications in images can deceive Florence-2's outputs.
- Ensuring robustness is critical for deployment in real-world applications (e.g., autonomous systems, security, medical AI).
- Vision transformers (like Florence-2) rely on patch-based tokenization, making them more vulnerable to small perturbations.

## Research Focus:

- Identify weaknesses in Florence-2's vision pipeline.
- Apply different adversarial attack strategies to test model stability.
- Test defensive mechanisms to improve robustness.
- Evaluate adversarial resistance across multiple datasets & tasks.

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# Thank you